



Explainable AI in Personalized Cancer Medicine

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March 2025



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- Master's students in Data Science
- EPFL, Switzerland

- CS-433 Machine Learning course project by Prof. Martin Jaggi
- Supervised by Prof. Jasmina Bogojeska (ZHAW)



OUTLINE

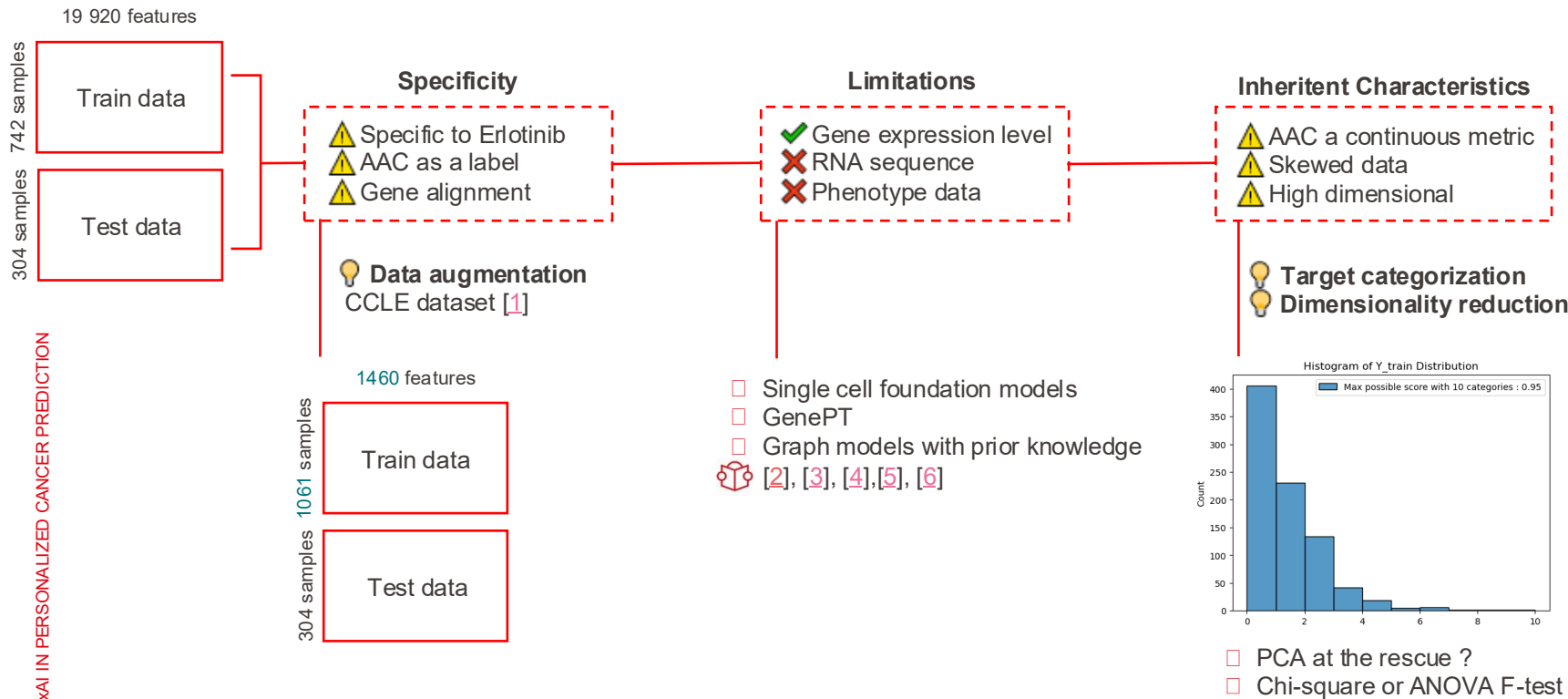
Dataset curation and models

Results

Explainability

Challenges and future work

Cancer genomic data for Erlotinib



Models

Base dataset

Augmented
dataset

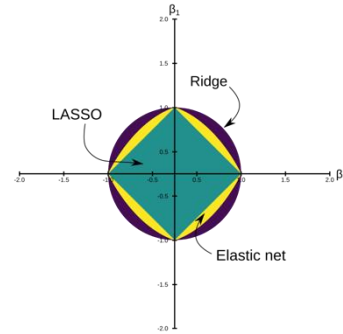
Categorized
targets and
reduced dataset

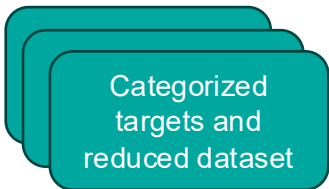
Categorized
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Linear models

- ☐ Linear regression
- ☐ Ridge / Lasso regression
- ☐ Elastic Net

Baselines





Linear models

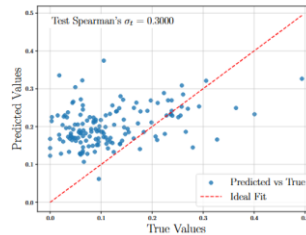
Support Vector Regressors (SVR) models

- ☐ Linear SVR
- ☐ Sigmoid SVR
- ☐ Polynomial / RBF SVR

} Kernel methods for
regression tasks

Tuned with grid search and cross validation

DRUG RESPONSE (AAC): PREDICTED VS TRUE VALUES
BEST LINEAR SVR WITH $\epsilon = 0.18$



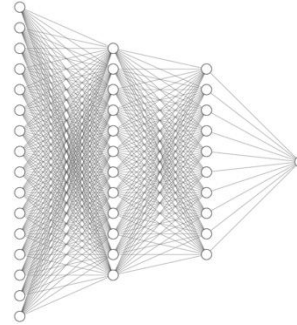
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Linear models

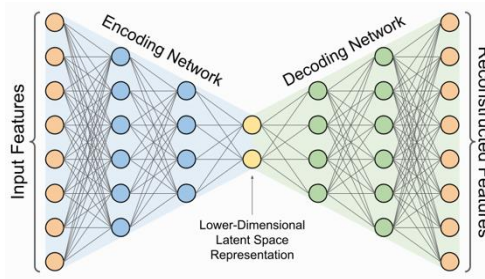
Support Vector Regressors (SVR) models

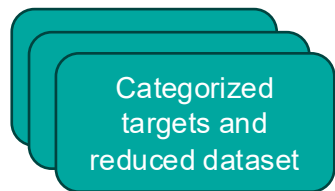
Neural Networks

- ☐ Neural networks
- ☐ Encoder Decoder models
- ☐ WPFS



- ☐ Stronger models that need more data
- ☐ How many layers for our **small** data?
- ☐ Data augmentation with noisy samples



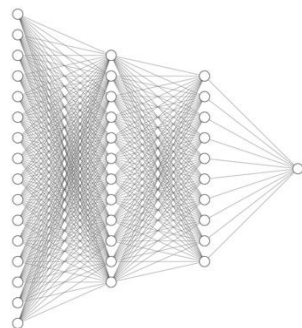
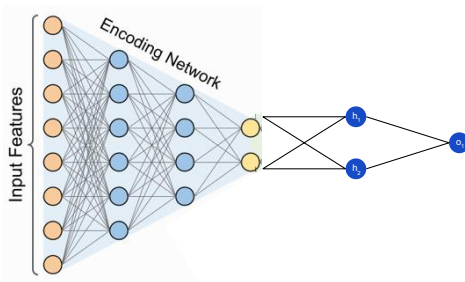


Linear models

Support Vector Regressors (SVR) models

Neural Networks

- ☐ Neural networks
- ☐ **Encoder Decoder models**
- ☐ WPFS



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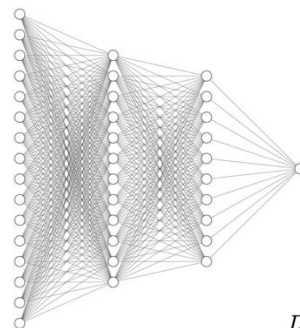
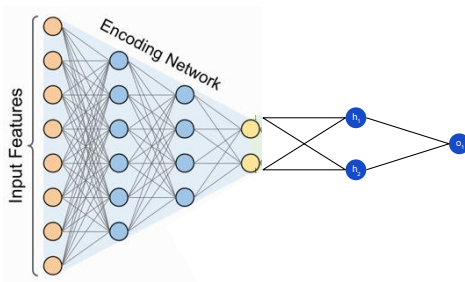
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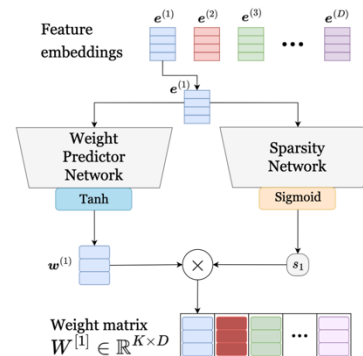
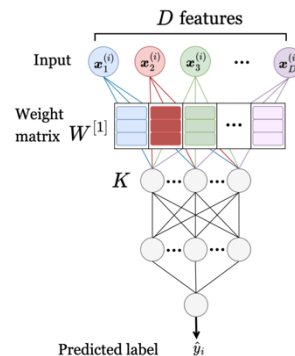
Support Vector Regressors (SVR) models

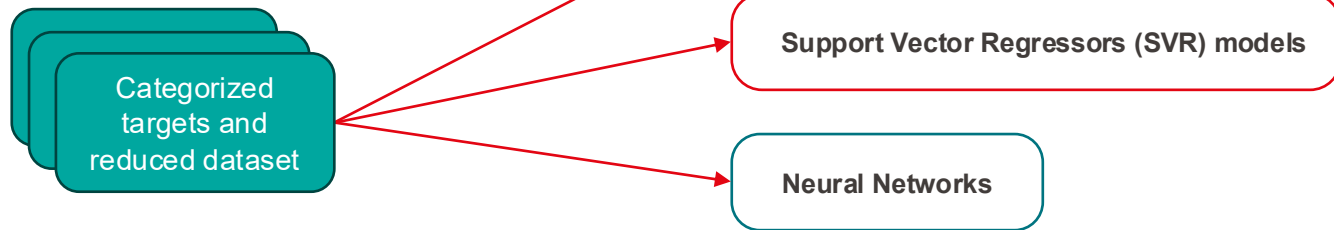
Neural Networks

- Neural networks
- Encoder Decoder models
- **WPFS**



- Stronger models that need more data
- How many layers for our **small** data?
- Data augmentation with noisy samples





Let's see how the models performed !



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Model	Kaggle Spearman's σ_k	Test Spearman's σ_t
Linear Regression	0.26	0.32
Elastic Net	0.41	0.39
Neural Network	0.39	0.61
Neural Network (w/ classes)	0.34	0.58
Encoder Decoder	0.16	0.21
WPFS	0.35	0.32
Linear SVR, $\epsilon = 0.18$	0.58	0.30
Linear SVR [*] , $\epsilon = 0.18$	-0.20	0.19
Sigmoid SVR, $\epsilon = 0.23$	0.56	0.30
Sigmoid SVR [*] , $\epsilon = 0.23$	0.53	0.36
Linear SVR, $\epsilon = 0.28, k = 500$	0.49	0.42

Models marked with ^{*} are trained on the augmented dataset.

TABLE I: RESULTING PERFORMANCE IN THE KAGGLE COMPETITION AND LOCALLY FOR THE BEST MODEL IN EACH CATEGORY.

Results: Linear Models



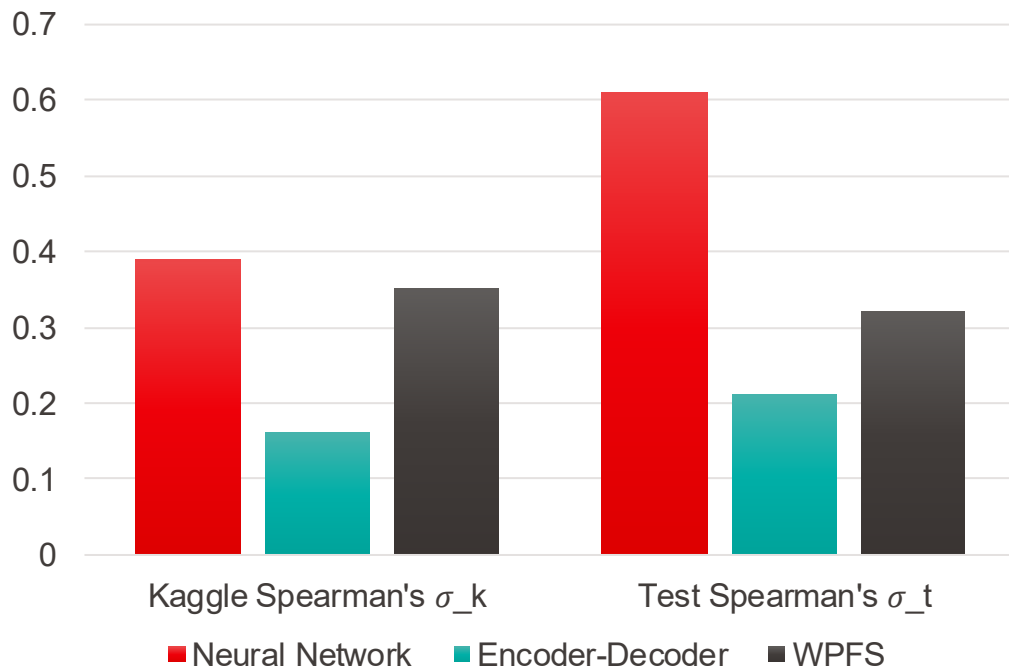
- Regularization effective for noisy data.
- Baseline with simple architecture.

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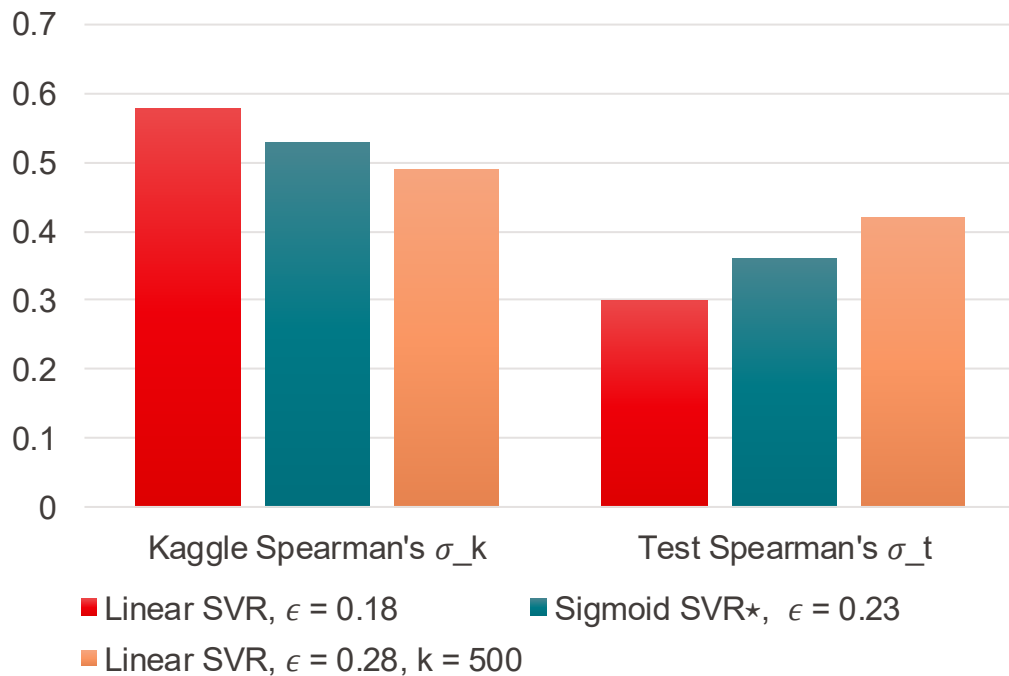
- Fast overfitting tendency.
- Deep models struggle with HDFS data.

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TABLE I: RESULTING PERFORMANCE IN THE KAGGLE COMPETITION AND LOCALLY FOR THE BEST MODEL IN EACH CATEGORY.

Results: Support Vector Regressors



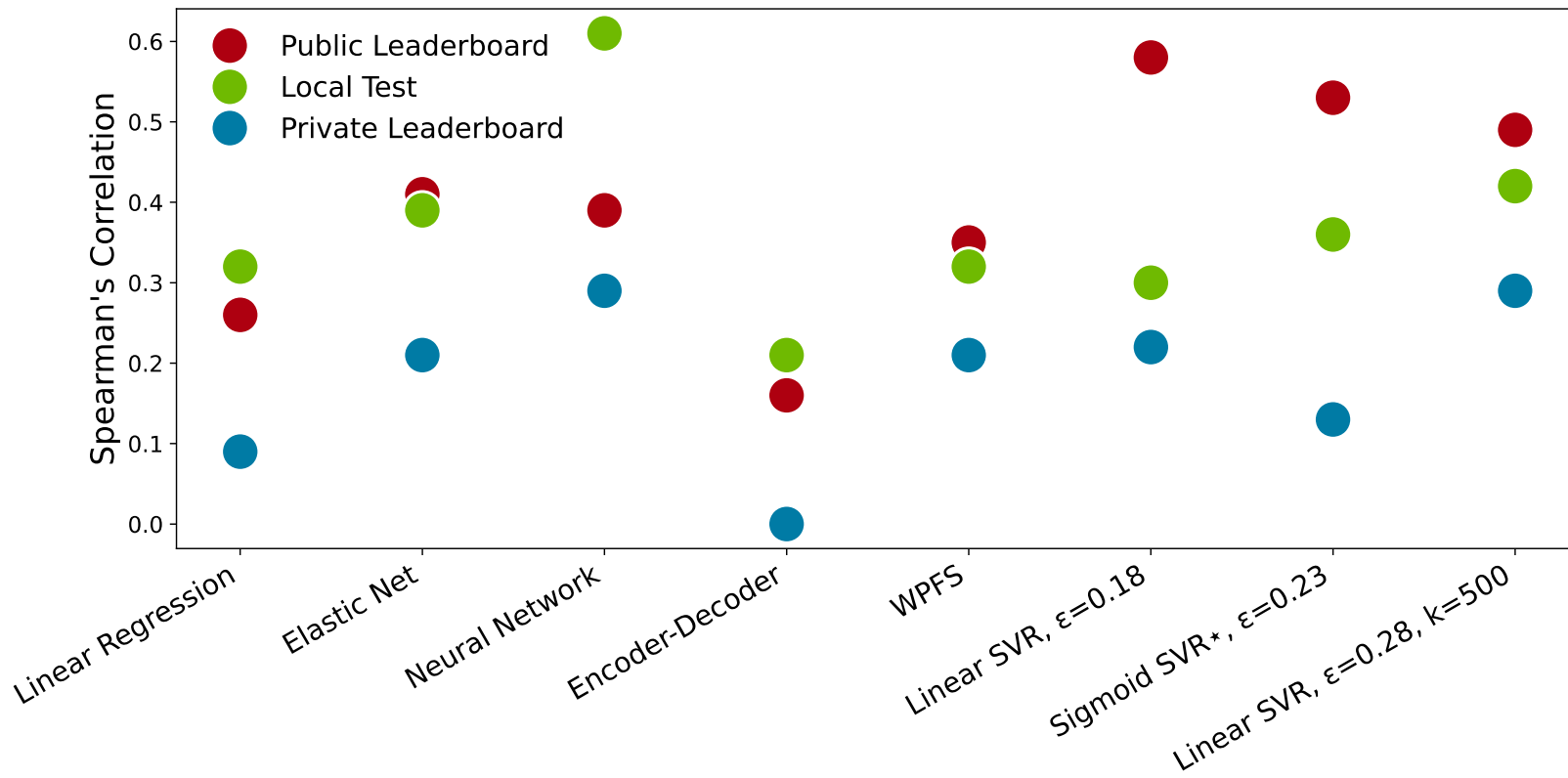
- Kernel-based methods are well suited for complex small-sized dataset.
- Data augmentation showed limited success.

Model	Kaggle Spearman's σ_k	Test Spearman's σ_t
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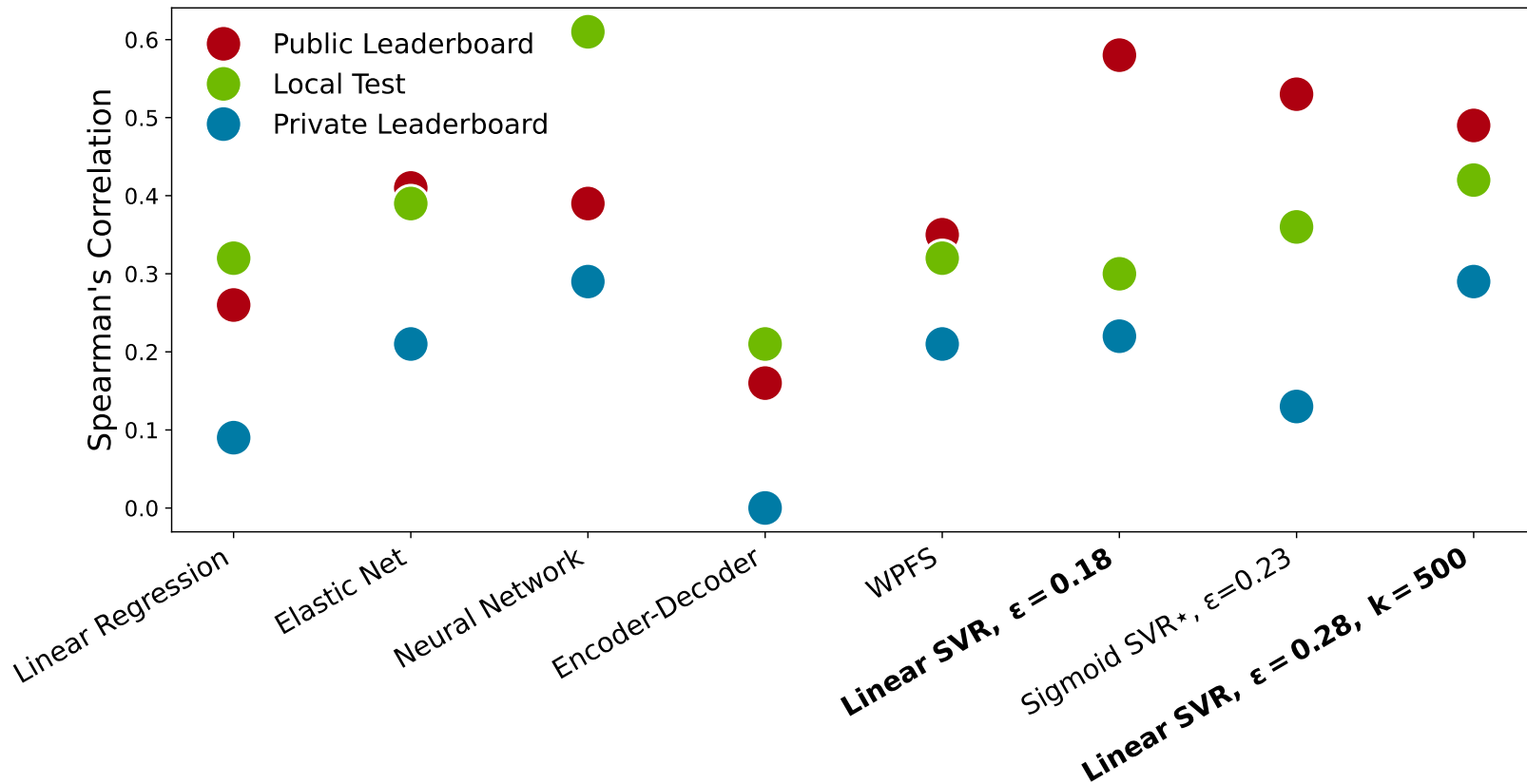
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TABLE I: RESULTING PERFORMANCE IN THE KAGGLE COMPETITION AND LOCALLY FOR THE BEST MODEL IN EACH CATEGORY.

Results: Summary



Results: Best Models



interpreting the model's output

OUTLINE

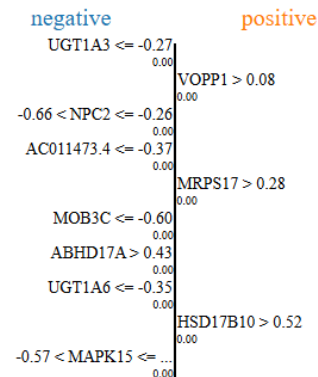
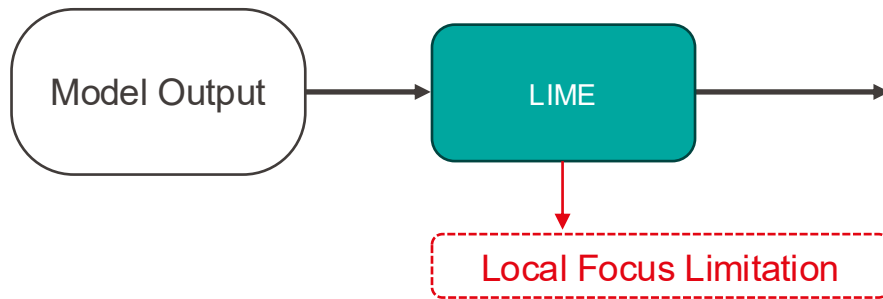
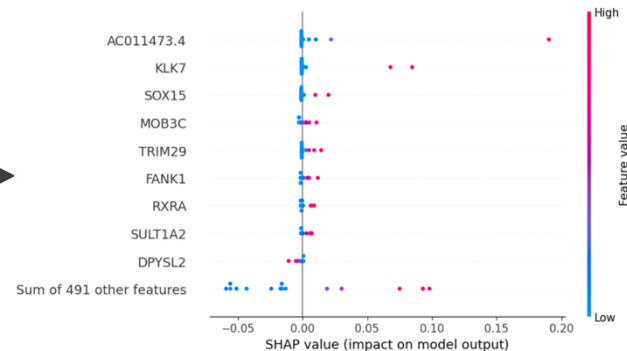
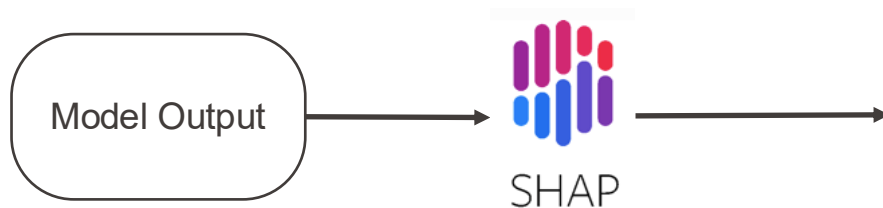
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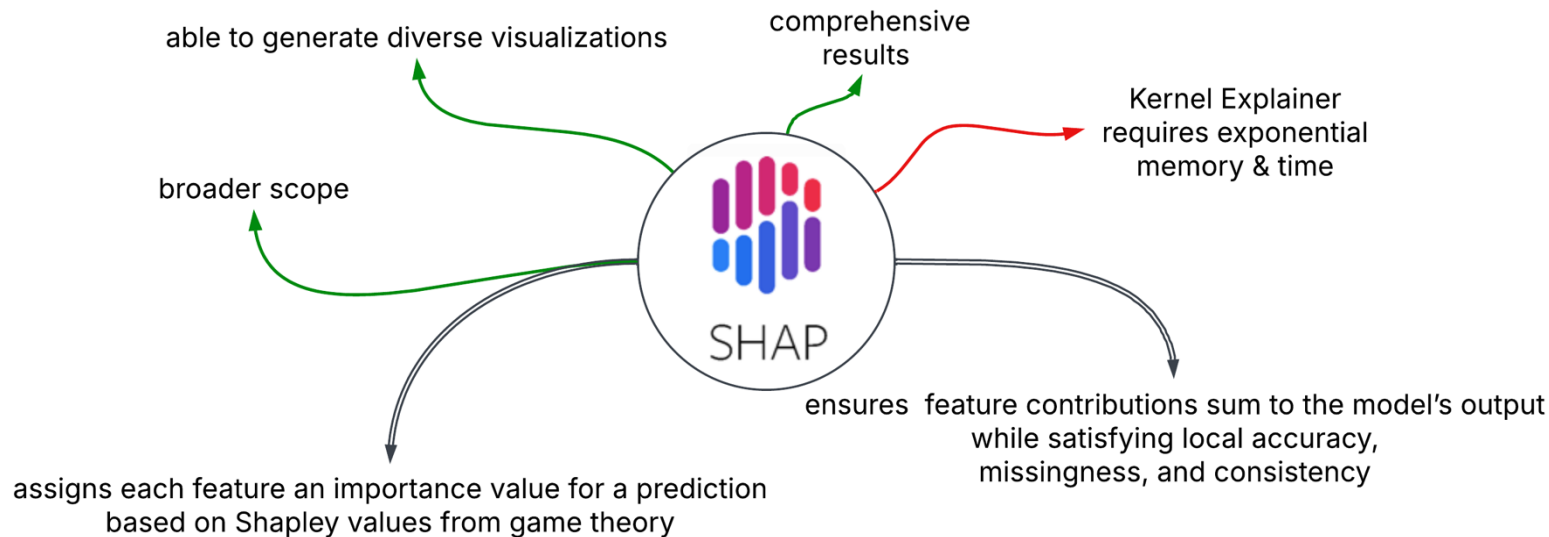
Two Approaches



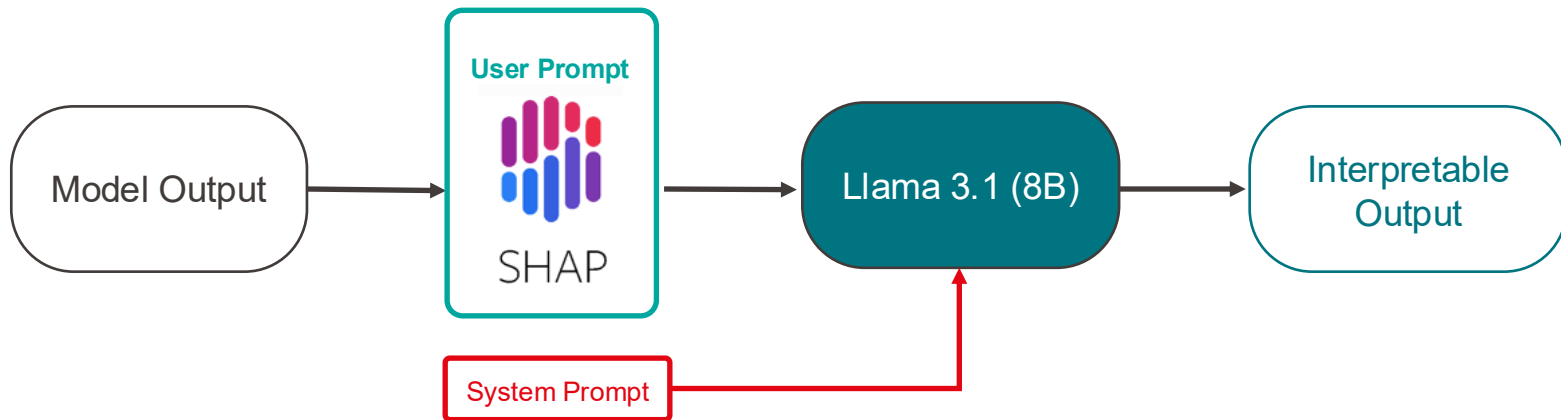
Feature	Value
UGT1A3	-0.27
VOPP1	0.73
NPC2	-0.48
AC011473.4	-0.37
MRPS17	0.82
MOB3C	-0.70
ABHD17A	1.19
UGT1A6	-0.35
HSD17B10	2.22
MAPK15	-0.56

More info available at : <https://shap.readthedocs.io/en/latest/index.html>
<https://github.com/marcotcr/lime/tree/master>

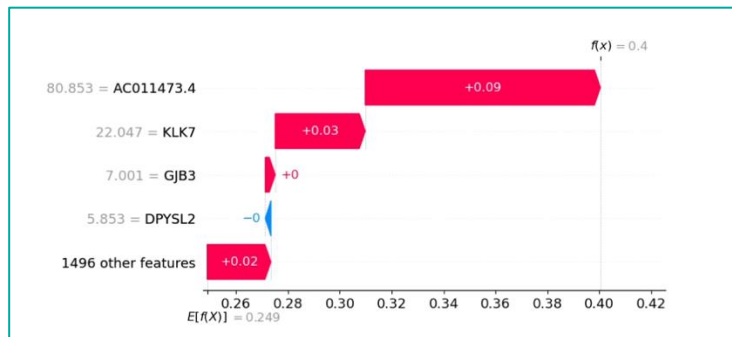
Why SHAP?



Final Automated Pipeline and Example



Concrete Example : TS28



Convert To Text

$f(x)$ = Model Output : **AAC**

System Prompt:

- **Behave** like a medical expert
- Use **complex** and **specialized** vocabulary
- **Formal** and **precise** answers required

User Prompt :

- **Task** : explain significance of top features

Feature	SHAP Value	Feature Value
AC011473.4	0.0906273	80.8526
KLK7	0.0345304	22.0465
GJB3	0.0036523	7.00119
DPYSL2	-0.00210705	5.8528

- Prediction : {**AAC**}
- Output **formatting**
- **Example** of output to follow (CoT manner)

Output Generation :

- **AC011473.4 (SHAP: 0.0906273)**: **Strong positive** contribution, suggesting its high expression (80.8526) plays a **key role** in drug sensitivity, potentially indicating a **favorable** response to erlotinib.
- **KLK7 (SHAP: 0.0345304)**: **Moderate positive** effect, with elevated expression (22.0465) **potentially influencing** tumor response, suggesting a possible synergistic interaction with erlotinib. [...]
- **DPYSL2 (SHAP: -0.00215705)**: **Slight negative influence**, suggesting reduced expression (5.8528) **marginally lowers** the AAC, indicating a **potential antagonist** effect on erlotinib response. [...]



OUTLINE

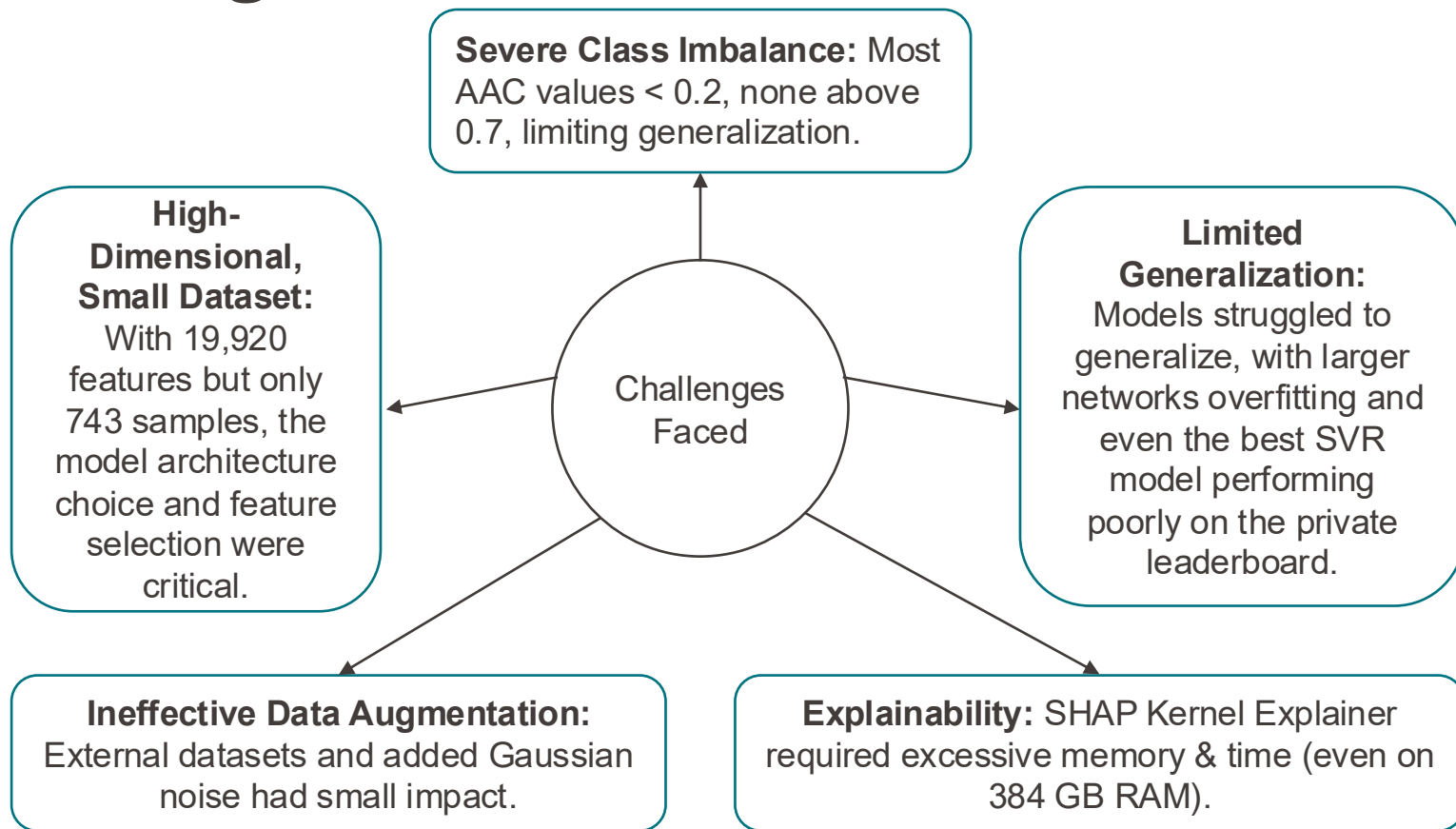
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Challenges Faced



- **Expert validation:** Work with clinicians and biologists to demonstrate the trustworthiness of the automatic tools (SHAP/LLaMA).
- **Curate more data** to improve model performances.
- **LLM Fine-Tuning & Alignment:** Improve SHAP result interpretation by fine-tuning on medical data (SFT) for greater domain knowledge, while applying preference alignment techniques (RLHF, DPO) to ensure results are presented in a user-preferred format.



Thank you



Questions

